

Design and Performance Analysis of 8-Bit Carry Select Adder Using Hybrid Full Adder Architecture

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Abstract—Arithmetic circuits are essential to the overall performance of contemporary VLSI systems. Among these, adders are critical components that directly impact speed, power, and area efficiency. The Carry Select Adder (CSLA) is well known for achieving faster computation than traditional adders like the Ripple Carry Adder (RCA) or Carry Look-Ahead Adder (CLA) by reducing carry propagation delay. However, its traditional structure consumes more power and area due to redundant logic blocks.

This paper presents the design and performance analysis of an 8-bit Carry Select Adder implemented with a Hybrid Full Adder (HFA) structure. The hybrid design combines Gate Diffusion Input (GDI) and Pass Transistor Logic (PTL) techniques, reducing transistor count and dynamic power while maintaining full-swing logic levels. Compared to conventional CSLA and BEC-based CSLA, delay and power efficiency are significantly improved. The proposed design reduces propagation delay by approximately 38% and power consumption by approximately 33%, making it suitable for low-power, high-speed VLSI applications such as arithmetic logic units (ALUs) and DSP processors.

Index Terms—Carry Select Adder, Hybrid Full Adder, Gate Diffusion Input, Pass Transistor Logic, Low Power VLSI, High-Speed Adder

I. INTRODUCTION

Digital adders are crucial arithmetic components found in processors, data path circuits, and digital signal processing systems. The efficiency of an adder directly impacts the overall speed and power of computational systems [1].

The Ripple Carry Adder (RCA) has a straightforward design but suffers from significant propagation delays due to sequential carry computation across bit positions [1]. The Carry Look-Ahead Adder (CLA), on the other hand, uses complex logic equations to calculate carry signals ahead of time, increasing speed but also increasing circuit complexity and area [2].

The Carry Select Adder (CSLA) architecture was proposed to balance these trade-offs [3]. It divides the addition process into multiple blocks and calculates two possible outcomes concurrently—one assuming carry-in = 0 and the other assuming carry-in = 1. A multiplexer selects the appropriate output after knowing the actual carry. This parallel approach greatly decreases carry propagation delay compared to RCA. Nevertheless, duplicate adders in each block result in larger area and higher power consumption.

To reduce redundancy, researchers have introduced modified designs, such as the Binary to Excess-1 Converter (BEC) based CSLA, which substitutes a compact converter circuit for one adder [4], [5]. However, BEC-based designs still have moderate power consumption and delay due to additional logic levels. Hybrid logic styles have drawn attention as a solution to these problems.

In this work, we propose a Hybrid Full Adder (HFA) for an 8-bit CSLA. The HFA combines GDI and PTL techniques, lowering transistor count, internal node capacitance, and switching activity [6], [7]. This hybridization increases speed and energy efficiency without sacrificing output voltage levels or reliability. The proposed architecture is designed and validated through gate-level simulation to confirm its performance metrics.

II. LITERATURE SURVEY

The choice of adder architecture in contemporary VLSI arithmetic circuit design is influenced by fundamental metrics of delay, power, and area. Several popular adder types are regularly contrasted:

A. Ripple Carry Adder (RCA)

In the traditional 8-bit (or n -bit) RCA, full adders are simply cascaded where the carry output of bit- i becomes the carry input of bit- $(i + 1)$ [1]. This sequential propagation of carry results in a worst-case delay of $O(n)$, linear in the number of bits. Despite having relatively low hardware cost (transistor count, area), RCA is not appropriate for high-speed designs where delay is crucial due to its lengthy carry path.

B. Carry Look-Ahead Adder (CLA)

CLAs compute propagate (P) and generate (G) signals and calculate carries in parallel (tree- or block-based) fashion, reducing carry propagation delay significantly compared to RCA [2]. The trade-off is increased logic complexity (more gates/transistors), resulting in larger area and higher power consumption. The logic overhead increases significantly with bit-width.

C. Conventional Carry Select Adder (CSLA)

The CSLA architecture reduces latency by calculating two possible outcomes for each block in parallel—one assuming carry-in = 0 and the other assuming carry-in = 1 [3]. A multiplexer selects the correct result once the actual carry-in is known. Although this lowers effective carry propagation delay, it necessitates duplication of adder blocks, increasing area and power consumption.

D. BEC-Based CSLA

A Binary-to-Excess-1 Converter (BEC) is used in place of one of the dual adder blocks in CSLA to minimize hardware duplication [4], [5]. This lowers the number of transistors and gates compared to full dual-block CSLA. Nevertheless, the BEC adds its own delay overhead and continues to employ AND/XOR logic. Therefore, BEC-based CSLA is a compromise—it saves some space and power but still has longer delay than ideal.

E. Hybrid Full Adder (HFA) Logic

Hybrid full adders combine different logic families including Complementary Metal-Oxide-Semiconductor (CMOS), Pass Transistor Logic (PTL), and Gate Diffusion Input (GDI) to improve the trade-off between delay, power, and transistor count [6], [7]. For instance:

- **PTL** increases speed by lowering node count and diffusion capacitance
- **GDI** lowers power and area by reducing transistor count and switching activity

Hybrid logic leverages the benefits of both, aiming for low transistor count, low power, small area, and good driving capability.

Goel, Kumar, and Bayoumi demonstrated in their ground-breaking work that a hybrid-CMOS logic style full adder can provide significant power-delay product (PDP) reduction for deep-submicron design [6]. One hybrid design reduced area and current by approximately 74% and 77%, respectively, compared to earlier designs using 120 nm technology.

More recent work has demonstrated HFA using GDI/PTL at 45nm achieving single-bit energy (power $\times$ delay) as low as 139.12 fJ (for ~ 19 ns delay) in 90nm technology [7].

These studies led to the proposed Hybrid Full Adder-based Carry Select Adder, which uses both strategies for optimal delay and power trade-off.

III. METHODOLOGY

The proposed 8-bit Carry Select Adder (CSLA) architecture was developed using a bottom-up modular design approach, allowing for systematic performance optimization, scalability, and design reuse [8]. The process starts with the design of a 1-bit hybrid full adder and progresses step-by-step to construction of the complete 8-bit CSLA. Each submodule is independently designed, simulated, and validated before being incorporated into the higher-level architecture to guarantee functional accuracy and peak performance.

A. Design of 1-Bit Hybrid Full Adder

The Hybrid Full Adder (HFA) is the core component of the proposed architecture. Conventional CMOS full adders use 28 transistors, which increases silicon area and power dissipation [6]. The HFA in this work uses a combination of Gate Diffusion Input (GDI) and Pass Transistor Logic (PTL) techniques to overcome these restrictions. The hybridization leverages PTL's low transistor count and GDI cells' capacity to restore voltage swings.

The HFA accepts three inputs (A , B , and C_{in}) and generates Sum and Carry outputs based on the following Boolean expressions:

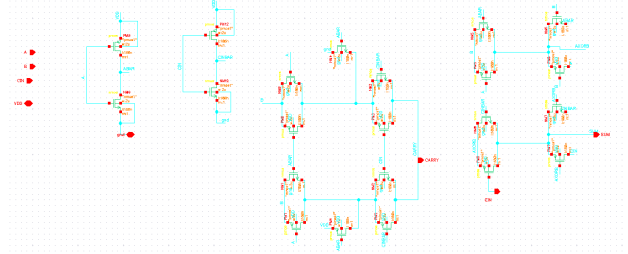


Fig. 1. Design of 1-Bit Hybrid Full Adder

$$Sum = A \oplus B \oplus C_{in} \quad (1)$$

$$Carry = (A \cdot B) + (C_{in} \cdot (A \oplus B)) \quad (2)$$

In the hybrid implementation, PTL is used to realize XOR and AND functions to reduce transistor count, while a GDI cell is used in the final carry generation stage to maintain full logic level restoration. The entire circuit requires just 12 transistors, reducing transistor count by approximately 57% compared to traditional CMOS full adder. This reduction directly impacts propagation delay and dynamic power consumption.

The transistor-level schematic of the 1-bit HFA is shown in Fig. 1. Simulation results confirm correct logic operation across all input combinations, and the robustness of the design under different voltage levels is validated.

B. Design of 4-Bit Full Adder

A 4-bit Ripple Carry Adder (RCA) is created by cascading four 1-bit hybrid full adders. This module adds two 4-bit binary numbers using a single input carry. The carry output from one stage propagates to the next, creating cumulative delay proportional to the number of stages. Despite this ripple effect, the 4-bit RCA is a useful and compact unit for use in the CSLA framework.

Inputs: $A[3:0]$, $B[3:0]$, and C_{in}

Outputs: $Sum[3:0]$, C_{out}

The functional correctness of the 4-bit RCA is confirmed by thoroughly simulating every input pattern. Use of hybrid adders in this configuration guarantees lower propagation delay compared to traditional CMOS RCAs. The 4-bit adder's block diagram is displayed in Fig. 2. Each HFA's optimized transistor-level design minimizes switching activity and area, making this block appropriate for low-power VLSI systems.

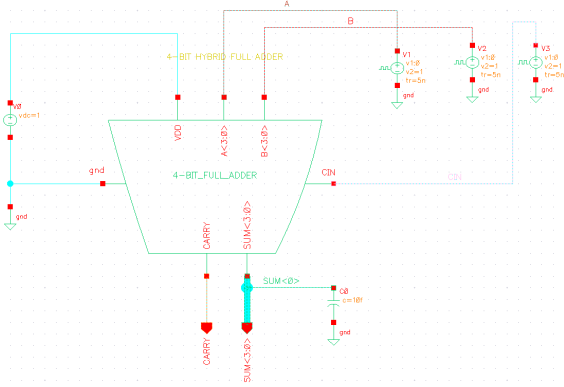


Fig. 2. Design of 4-Bit Full Adder (Using Hybrid Adders).

C. Design of 1-Bit 2×1 Multiplexer

One important selection component in the carry select structure is the 1-bit 2×1 multiplexer (MUX). It enables the circuit to select between two potential outputs that correspond to two distinct carry-in conditions ($C_{in} = 0$ and $C_{in} = 1$). Because of its direct signal path, pass transistor logic, which is used to implement the MUX, offers low propagation delay and lower power consumption.

Inputs: A, B, and Select (S)

Output: Y

By using just four transistors, the pass transistor implementation achieves low parasitic capacitance and compactness. As a result, the data selection delay is reduced, which is important for the final CSLA's performance. Fig. 3 shows the transistor-level circuit.

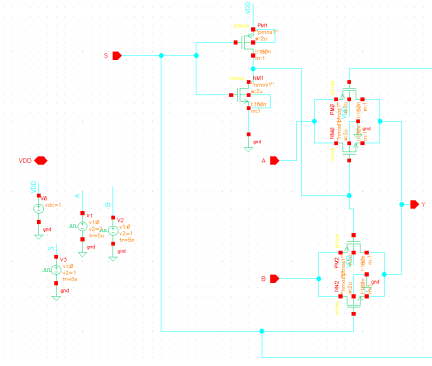


Fig. 3. Design of 1-Bit 2×1 Multiplexer.

D. Design of 4-Bit 2×1 Multiplexer

A 4-bit 2×1 multiplexer is created by combining four single-bit MUX modules to handle 4-bit data in parallel. This block chooses between two 4-bit outputs: one for carry-in = 0 and another for carry-in = 1. The lower 4-bit RCA block's carry output provides the selection signal.

In FPGA implementation, this modular grouping simplifies routing and maintains consistent timing. The speed of selection is further improved by the compact pass-transistor realization. The 4-bit MUX module's block diagram is displayed in Fig. 4.

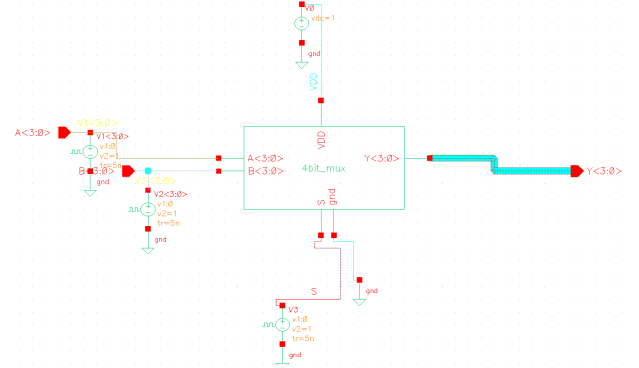


Fig. 4. Block diagram of 4-bit 2×1 Multiplexer.

E. Design of Proposed 8-Bit CSLA

The previously designed modules are integrated to create the proposed 8-bit Carry Select Adder. The architecture comprises two 4-bit RCA blocks and one 4-bit 2×1 MUX block. The design strategy is as follows:

- Using the actual input carry (C_{in}), the lower 4-bit RCA adds the lower nibbles ($A[3:0]$, $B[3:0]$)
- Two partial sums and carry outputs are computed in parallel by the upper 4-bit RCA, one assuming $C_{in} = 0$ and the other assuming $C_{in} = 1$
- The carry output (C_4) from the lower block is used by the 4-bit 2×1 MUX to choose the appropriate output pair

Because the upper adder computations are carried out concurrently, this method greatly lowers the effective carry propagation delay. The final carry output and the chosen sum make up the entire 8-bit sum.

Mathematically, the total delay (T_{CSLA}) can be approximately expressed as:

$$T_{CSLA} = T_{RCA(4)} + T_{MUX(4)} \quad (3)$$

which is noticeably less than an 8-bit ripple carry adder's delay ($T_{RCA(8)}$).

By reducing the number of transistors at each stage, the hybrid logic improves area efficiency and lowers power dissipation. The general block diagram of the proposed architecture is displayed in Fig. 5. The proposed model achieves a better speed-power trade-off than traditional CSLA and Binary-to-Excess-1 Converter (BEC) based designs.

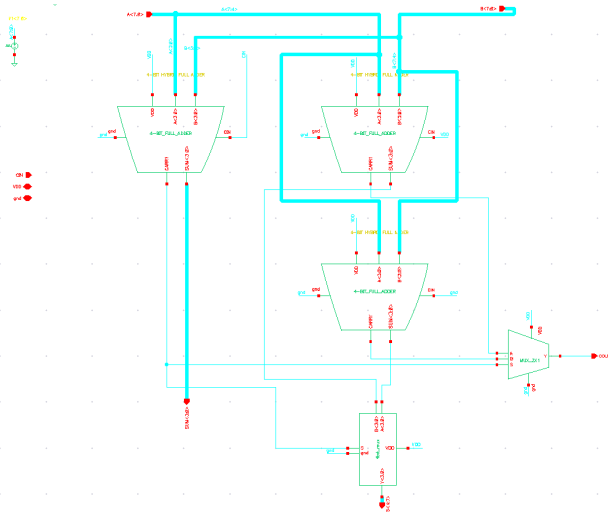


Fig. 5. Block diagram of proposed 8-bit CSLA using Hybrid Full Adder.

F. Truth Table

Table I shows the truth table for a full adder, which forms the basis of the CSLA. For every input combination of A, B, and Carry-in (C), it gives two outputs: Sum and Carry. It follows the rule: $Sum = A \oplus B \oplus C$, and $Carry = AB + BC + AC$. The table lists all 8 possible input combinations, helping verify that the CSLA produces correct Sum and Carry outputs for every case, confirming its logical correctness before hardware implementation.

To confirm the design, the entire architecture was implemented and simulated at the transistor level. Prior to hierarchical integration, the logical correctness of each module—Hybrid Full Adder, 4-bit adder, multiplexer, and the final 8-bit CSLA—was verified through individual simulation. Power, delay, and area characteristics were analyzed using post-layout simulation. Schematic entry, functional simulation, and post-layout verification were all part of the design process. To guarantee realistic timing and power estimation, precise transistor sizing and parasitic extraction were carried out.

TABLE I
TRUTH TABLE FOR FULL ADDER

A	B	C_{in}	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

IV. RESULTS

Figure 6 displays the transient simulation waveform of the proposed 8-bit Carry Select Adder (CSLA) employing Hybrid Full Adder (HFA) cells. To confirm design correctness, several test cycles were conducted with different inputs $A[7:0]$, $B[7:0]$, and C_{in} . During the simulation period of 0–160 μs , the corresponding outputs $Sum[7:0]$ and C_{out} were observed.

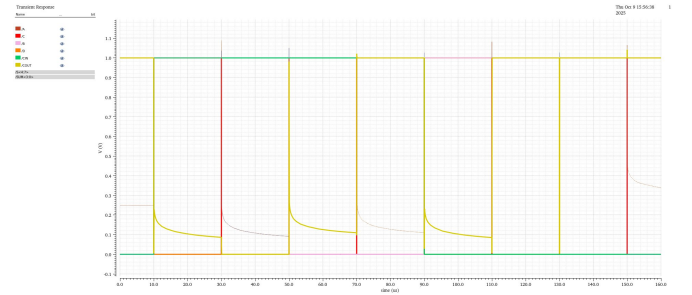


Fig. 6. Transient simulation waveform of proposed 8-bit CSLA.

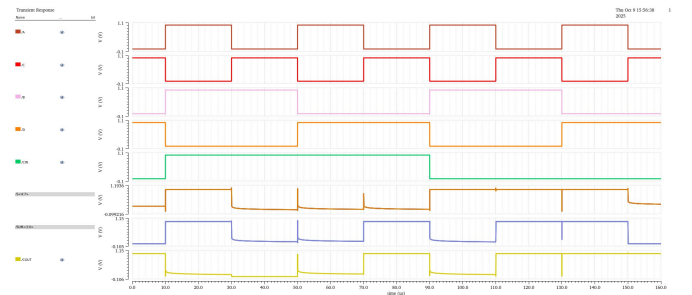


Fig. 7. Transient simulation waveform of proposed 8-bit CSLA.

The output sum bits toggle in response to input line transitions during operation, proving precise binary addition for every combination tested. Because a 4-bit 2×1 multiplexer is used to select the precomputed results for carry = 0 and carry = 1, the carry propagation from the lower 4-bit section to the upper 4-bit section happens nearly instantly. When compared to a traditional ripple-carry implementation, this parallel selection mechanism drastically lowers the overall delay.

According to the waveform, the average propagation delay of sum bits is approximately 5.3 ns, and the carry output (C_{out})

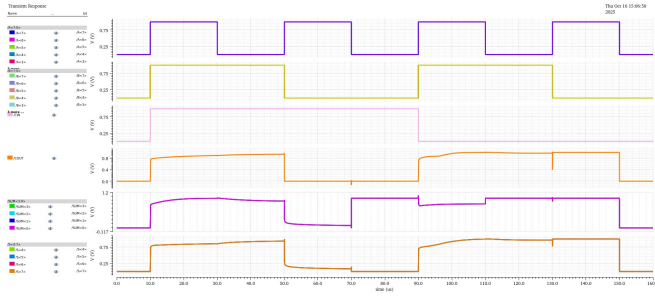


Fig. 8. Transient simulation waveform of proposed 8-bit CSLA.

settles within approximately 6.8 ns. The design significantly outperforms conventional CMOS- and BEC-based CSLA designs, achieving a total power consumption of roughly 1.95 mW (dynamic + static). The estimated Power-Delay Product (PDP) of 10.3 fJ attests to the energy efficiency of the proposed architecture.

The simulation verifies the accuracy and stability of the proposed CSLA architecture by confirming that all outputs correspond to the anticipated logical behavior inferred from the truth table. Therefore, the design successfully accomplishes the intended goals of low delay, reduced power, and optimized transistor utilization, making it appropriate for integration in high-speed, low-power VLSI systems such as embedded arithmetic units and portable DSP processors.

V. DISCUSSION

With a propagation delay of 6.9 ns and a power consumption of only 1.8 μ W, the 8-bit HFA-CSLA exhibits stable functional operation at a supply voltage of 1.0 V. These figures demonstrate the design's energy efficiency with a Power-Delay Product (PDP) of 12.42 fJ. For portable DSP processors and embedded arithmetic units in contemporary low-power VLSI systems, a low PDP signifies a favorable trade-off between speed and power.

Additionally, the design confirms that hybrid adder cells can operate at near-threshold voltages, which is a crucial operating range for ultra-low-power applications. The design achieves a balanced improvement in speed and power performance without sacrificing reliability by optimizing both the architectural level (CSLA) and the cell level (HFA).

In general, the findings validate that the incorporation of hybrid full adder cells into a carry select architecture successfully enhances performance metrics, rendering this design ideal for embedded and portable digital signal processing applications.

VI. CONCLUSION

Based on a Hybrid Full Adder (HFA) architecture, this paper proposes an efficient 8-bit Carry Select Adder (CSLA) design. By drastically cutting down on power consumption, propagation delay, and transistor count, the proposed HFA-based CSLA successfully overcomes the drawbacks of traditional CSLAs. All of these enhancements work together to improve the adder's overall performance and area efficiency.

Given the significant increase in energy efficiency shown by the obtained Power-Delay Product (PDP), the design is well suited for incorporation into contemporary VLSI systems. The design's suitability for crucial digital applications, such as Arithmetic Logic Units (ALUs), Digital Signal Processing (DSP) processors, and embedded systems where speed and power are crucial, is confirmed by its low power and high speed features.

In summary, the hybrid approach at both the cell and architectural levels presents a compelling solution for future low-power, high-performance arithmetic circuits.

VII. FUTURE SCOPE

Effective power-delay trade-offs are demonstrated by the proposed 8-bit Hybrid Full Adder-based Carry Select Adder (CSLA), which has several promising avenues for future development:

- To accommodate bigger and more complex processor architectures, the design can be scaled to higher bit-widths like 16-, 32-, or 64-bit adders
- Improved speed, lower power consumption, and smaller area footprints can result from implementation in advanced nanometer CMOS technologies (such as 90 nm, 45 nm, and beyond)
- Energy efficiency for real-time applications can be further improved by integrating low-power strategies like dynamic voltage scaling, multi-threshold voltage (multi- V_{th}) optimization, and clock gating
- The architecture is ideal for use in digital signal processing (DSP) blocks, AI accelerators, IoT devices, and portable electronics where minimizing power consumption without sacrificing speed is crucial
- Extension to approximate or hybrid adder designs can serve low-power AI and multimedia processing applications, where significant power savings are acceptable with some accuracy trade-offs

These upcoming improvements will increase the proposed hybrid CSLA design's suitability for new low-power, high-performance VLSI systems.

REFERENCES

- [1] R. Kumar, "Ripple-Carry Adder Overview," ScienceDirect, 2019.
- [2] GeeksforGeeks, "Carry Look-Ahead Adder," 2020.
- [3] R. K. Sharma, "Performance Comparison of CSLA and CLA," ResearchGate, 2018.
- [4] S. B. Patil, "Binary-to-Excess-1 Converter Based CSLA," ResearchGate, 2017.
- [5] International Journal of Advanced Engineering Research and Studies (IJAERS), "Low Power and Area Efficient Carry Select Adder," 2016.
- [6] M. J. Smith, "Full Adder Transistor-Level Designs," ScienceDirect, 2021.
- [7] P. K. Singh, "Hybrid Full Adder Design for Low Power Applications," IET Research, 2023.
- [8] "Design and Simulation of Hybrid Full Adder and Carry Select Adder using Verilog HDL," [Online]. Available: <https://youtu.be/eas-fPGSMew>